Hostel Management System

Unit Test Report

Testing phase: unit test results for the Hostel Management System

Project	Hostel Management System (HMS)
Team	Team 3
Authors	Meghana P (PES2UG24CS642) B S Sharath Chandra Reddy (PES2UG24CS630) Adarsha V (PES2UG24CS621) Mohammadnoor Sirasgi (PES2UG25CS812)
Repository	github.com/adarshv15/hostel-management-system-team3 (commit 857e9c1)
Baseline document	Software Requirements Specification v1.0 (TEAM_3_HMS_SRS)
Report version / date	1.0 / 09-11-2026
Status	Draft for team review
Test period	Testing phase (plan: 2 Nov to 6 Nov); this run executed 09-11-2026

1. Introduction

This report records the results of the unit testing carried out on the Hostel Management System (HMS) for the Testing phase (step 2a and 2b of the project plan). It covers the automated Django test suite in `hostel/tests.py`, run against the code at commit `857e9c1`.

The suite exercises the model-level business rules (allocation capacity, payment limits, date validation, uniqueness constraints), the admin audit trail, and the dashboard and health endpoints. Requirement references use the IDs from the Software Requirements Specification v1.0.

1.1 References

- Software Requirements Specification v1.0 (Team 3), sections 4 and 5.
- Repository documents `docs/TEST_CASES.md`, `TESTING_DOCUMENTATION.md` and `docs/SPRINT_2.md`.
- The Software Test Plan was not available in the repository, so test IDs below (UT-01 to UT-32) are assigned in this report and are not the Test Plan's own identifiers.

2. Test Environment

Item	Detail
Operating system	Linux 6.18 (cloud sandbox)
Python	3.13.18 (the project CI targets Python 3.12)
Django	5.2.19 (requirements.txt allows >=5.1,<6.0)
Coverage tool	coverage.py 7.17.2 with `--branch`
Database	SQLite, Django-managed in-memory test database
Test runner	Django `manage.py test` (unittest based, `django.test.TestCase`)
Execution date	09-11-2026

Commands executed from the repository root:

```
python manage.py check
python manage.py test -v 2
```

3. Summary of Results

Measure	Result
Test cases designed	32
Test cases executed	32
Passed	32
Failed / errors / skipped	0 / 0 / 0
Pass rate	100%
Run time	4.447 s (test runner total)
Django system check	Passed: no issues (0 silenced)

Overall verdict: PASS. All 32 unit tests passed on the first run with no failures, errors or skipped tests. This agrees with the 32-test result recorded in `TESTING_DOCUMENTATION.md` on 29-10-2026.

3.1 Results by functional area

Area	Tests	Passed	Failed	Pass rate
Platform	3	3	0	100%
Notices	2	2	0	100%
Student	1	1	0	100%
Room & Allocation	7	7	0	100%
Fees & Payments	6	6	0	100%
Complaints	1	1	0	100%
Leave Requests	2	2	0	100%
Visitor Logs	4	4	0	100%
Meal Attendance	2	2	0	100%
Audit Logging	4	4	0	100%
Total	**32**	**32**	**0**	**100%**

4. Detailed Test Case Results

Each row is one test method in `hostel.tests.HostelModelTests`. The Result column is taken from the verbose runner output (`... ok`).

ID	Test method	What it verifies	SRS reference	Result
UT-01	`test_django_configuration`	Django system check reports no configuration issues.	Supporting	**Pass**
UT-02	`test_home_page_returns_dashboard`	Dashboard page `/` returns HTTP 200 and shows the system title and student records section.	SRS 2.2 dashboards	**Pass**
UT-03	`test_dashboard_only_displays_published_notices`	Published notices appear on the dashboard; draft notices do not.	HMS-F-050	**Pass**
UT-04	`test_health_endpoint`	`/health/` returns HTTP 200 with status `ok` JSON.	Supporting	**Pass**

ID	Test method	What it verifies	SRS reference	Result
UT-05	`test_create_student_record`	A student record (ID, name, email, phone, status) is stored and its string form is correct.	HMS-F-001 (partial)	**Pass**
UT-06	`test_room_starts_with_capacity_available`	A new room starts with all beds available.	HMS-F-010	**Pass**
UT-07	`test_create_room_record`	A room with block, number, type and capacity is stored with the correct label.	HMS-F-010	**Pass**
UT-08	`test_allocation_uses_room_capacity`	Allocating the only bed in a room leaves zero available beds.	HMS-F-011	**Pass**
UT-09	`test_database_prevents_double_allocation_of_same_bed`	The database constraint rejects a second active allocation of the same bed.	HMS-F-012	**Pass**
UT-10	`test_create_invoice`	A fee invoice can be created for a student with amount and due date.	HMS-F-021 (partial)	**Pass**
UT-11	`test_add_payment_record`	A payment (mode, reference, recorder) can be recorded against an invoice.	HMS-F-023	**Pass**
UT-12	`test_fee_balance_calculation`	Outstanding balance is correct across multiple payments.	HMS-F-023	**Pass**
UT-13	`test_create_update_complaint`	A complaint is created as Open, then assigned to a staff user and moved to Resolved.	HMS-F-030, F-031	**Pass**
UT-14	`test_create_leave_request`	A leave request with dates and reason is stored.	HMS-F-041	**Pass**
UT-15	`test_leave_end_date_cannot_precede_start_date`	Validation rejects an end date earlier than the start date.	HMS-F-041	**Pass**
UT-16	`test_add_visitor_entry`	A visitor entry is stored with timestamp and host student.	HMS-F-040	**Pass**
UT-17	`test_record_visitor_exit_time`	A visitor exit timestamp is persisted.	HMS-F-040	**Pass**
UT-18	`test_record_meal_attendance`	Attendance is recorded per student, meal and date.	HMS-F-042	**Pass**
UT-19	`test_meal_attendance_prevents_duplicate_student_meal_date`	A duplicate student/meal/date entry is rejected by the database.	HMS-F-042	**Pass**
UT-20	`test_create_notice`	A notice can be created by an authenticated user.	HMS-F-050	**Pass**
UT-21	`test_audit_event_logged`	An audit event record can be stored.	HMS-SR-005 (partial)	**Pass**
UT-22	`test_admin_student_creation_is_audited`	Creating a student in the admin writes an audit event with the actor.	HMS-SR-005 (partial)	**Pass**
UT-23	`test_admin_student_update_is_audited`	Updating a student in the admin writes an audit event.	HMS-SR-005 (partial)	**Pass**
UT-24	`test_admin_bulk_delete_creates_audit_events`	Bulk deletion in the admin writes one audit event per record.	HMS-SR-005 (partial)	**Pass**
UT-25	`test_new_allocation_exceeding_room_capacity_with_different_bed_label`	A new allocation beyond capacity is rejected even with a different bed label.	HMS-F-012	**Pass**
UT-26	`test_moving_allocation_into_full_room_is_rejected`	Moving an allocation into a full room is rejected.	HMS-F-012	**Pass**
UT-27	`test_updating_allocation_within_current_full_room_succeeds`	Editing an allocation inside its own full room is still permitted.	HMS-F-012	**Pass**
UT-28	`test_payment_amount_zero_or_less_rejected`	Payments of zero or a negative amount are rejected.	HMS-F-023	**Pass**

ID	Test method	What it verifies	SRS reference	Result
UT-29	`test_payment_exceeding_remaining_invoice_balance_rejected`	A payment larger than the outstanding balance is rejected.	HMS-F-023	**Pass**
UT-30	`test_payment_exactly_equal_to_remaining_balance_succeeds`	A payment equal to the outstanding balance is accepted.	HMS-F-023	**Pass**
UT-31	`test_visitor_exit_time_earlier_than_entry_time_rejected`	Validation rejects an exit time earlier than the entry time.	HMS-F-040	**Pass**
UT-32	`test_visitor_valid_exit_time_and_missing_exit_time_succeed`	A valid exit time and an ongoing visit (no exit) are both accepted.	HMS-F-040	**Pass**

5. Observations and Defects

No test failed. The following issues were found while analysing the code and the coverage data. They are gaps in the tested behaviour or in the implementation, not failing tests.

#	Observation	Severity	Recommended action
O-1	Audit signal handlers in `hostel/signals.py` are never registered: `hostel/apps.py` has no `ready()` method, so the module is never imported (0 of 17 statements executed). Audit events exist only for changes made through the Django admin. A direct ORM write was confirmed to create no audit event.	Medium	Either register the module in `HostelConfig.ready()` (and avoid double logging with `AuditLoggingAdmin`) or delete the unused module.
O-2	The rule that only ACTIVE students can be allocated a bed (`Allocation.clean`, `models.py` line 62) has no unit test. A manual probe confirmed the rule works (a pending student was rejected).	Medium	Add a test that a PENDING, REJECTED or INACTIVE student is rejected.
O-3	`FeeInvoiceAdmin` balance columns (`total_paid`, `balance_due`, annotated queryset) are never executed by any test (`admin.py` lines 152-174).	Medium	Add an admin changelist test that checks the displayed paid and balance amounts.
O-4	Single-record deletion through the admin (`delete_model`, `admin.py` lines 45-58) is not tested, although bulk deletion is.	Low	Add a test for individual delete producing one audit event.
O-5	Model `clean()` rules run only through admin and ModelForm workflows. Direct `objects.create()` or `.save()` skips them (already documented by the team).	Low	Keep documented; call `full_clean()` in any future service or API code.
O-6	`FeeInvoice.status` does not change automatically when payments are recorded (already documented by the team).	Low	Track as a gap against HMS-F-023 and HMS-F-022.

6. Limitations

- The tests use Django's test client and a temporary in-memory database. They do not test a deployed environment, a real browser or real user accounts.
- The suite is model and endpoint level. There are no tests for requirements that are not yet implemented (for example document upload, room transfer requests, payment gateway, reminders and notifications).
- This run used Python 3.13.18; the project's CI uses Python 3.12. Results were identical to the previously recorded Python 3.12 run in test count and outcome.
- Manual checks reported earlier by the project owner (dashboard access, admin login, student creation, audit logging, room creation and allocation) were not repeated in this run.

7. Conclusion

The unit test phase is complete for the functionality that has been built: 32 of 32 tests pass. Coverage analysis (see the Code Coverage and Branch Coverage reports) shows that the remaining untested areas are small and are listed in section 5. Closing observations O-1 to O-3 would give the most benefit.

Sign-off

Role	Name	Signature	Date
Prepared by			
Reviewed by			
Approved by (Project Guide)			

Appendix A. Runner Output (excerpt)

```
Found 32 test(s).
System check identified no issues (0 silenced).
test_add_payment_record ... ok
test_add_visitor_entry ... ok
test_admin_bulk_delete_creates_audit_events ... ok
test_admin_student_creation_is_audited ... ok
test_admin_student_update_is_audited ... ok
   ... (26 further tests, all ok) ...
test_visitor_valid_exit_time_and_missing_exit_time_succeed ... ok

----------------------------------------------------------------------
Ran 32 tests in 4.447s

OK
```